

Privacy Masker: PII Detection and Masking in Mobile UI Screenshots

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Abstract—Mobile applications routinely expose sensitive personal information (PII) such as email addresses, account balances, and names. This creates a privacy hazard when users share screenshots. We present Privacy Masker, a deep learning pipeline that automatically detects PII regions in Android UI screenshots and produces pixel-precise bounding boxes for downstream redaction. We explore two architectures: (1) a single-stage Vision-Language Model (PaliGemma 2 with LoRA) that frames detection as autoregressive sequence generation, achieving $F1 = 0.449$; and (2) a two-stage pipeline (PaddleOCR with a Gemma-4 LoRA classifier) that decouples localization from classification, reaching $F1 = 0.683$ on the pii_v1 dataset and 0.586 on the larger pii_v5 dataset. Our results show that task decomposition with layout-aware prompting outperforms end-to-end VLM fine-tuning at this data scale.

Index Terms—PII detection, vision-language model, LoRA, OCR, mobile UI, bounding box detection

I. INTRODUCTION

Mobile applications routinely expose sensitive personal information (PII) such as emails, account balances, and names. This creates a privacy hazard when users share screenshots, since information the sender never intended to expose may be visible in the background. Privacy Masker addresses this by automatically detecting PII regions and producing bounding boxes for downstream redaction.

PII detection requires visual layout understanding, semantic context (distinguishing a person’s name from a UI heading), and pixel-precise localization. Pure regex or OCR approaches fail on unstructured PII, making Vision-Language Models (VLMs) a natural fit for the task.

II. DATA ENGINEERING

A. Source Dataset

Both datasets were constructed from the RICO-ScreenQA corpus [1], [2], a collection of real Android UI screenshots with pixel-precise bounding boxes and natural-language question–answer pairs.

B. Dataset Versions

We built two curated datasets. The first, pii_v1, contains 3,645 screens with broad keyword regex matching and no stratified split. It was used for the single-stage VLM experiments (Section III) and for the two-stage pipeline iterations v1 and v2 (Section IV). The second, pii_v5, contains 9,989 screens with

a precision-first regex engine, stratified multi-label splitting via the rarest-label heuristic, pruned ambiguous labels, and biometrics added under the other_sensitive category. It was used for two-stage pipeline iteration v3.

Both datasets follow the same per-screen schema. Each line in a JSONL file contains an image path, image dimensions, a screen ID, and a list of objects, each consisting of a bounding box $[x_{\min}, y_{\min}, x_{\max}, y_{\max}]$ in absolute pixels and a PII label. Screens with no PII contain an empty object list and serve as hard negatives, comprising roughly 12–15% of each split.

C. Label Taxonomy

Nine PII categories are defined, as shown in Table I.

TABLE I
PII LABEL TAXONOMY

Label	v5 Train
address	3,135
transaction_amount	2,316
email_address	1,975
other_sensitive	1,260
date_of_birth	884
phone_number	816
username	570
full_name	201
account_balance	153

D. Precision-First Regex Engine

QA pairs were classified using a compiled keyword and regex map applied to the question text. Candidate keywords were identified by performing n-gram frequency analysis across both question and answer texts - this allowed identifying which phrases correlate with PII content. High-noise keywords such as the bare terms *name* and *date*, which caused over 40% false positives in earlier versions, were removed in favour of structurally unambiguous multi-word patterns such as *date of birth* and *first name*. Several labels present in earlier dataset versions were removed following semantic analysis of the source QA data, as they lacked reliable structural patterns in question phrasing necessary for precise regex matching (e.g. *id_number*, *account_number*). Fig. 1 shows an example screen before and after PII annotation.

E. Stratified Splitting via Rarest-Label Heuristic

Each screen is assigned a primary label equal to the globally rarest PII label it contains. Screens are then stratified by primary label and split 80/10/10 by screen ID, ensuring that rare categories such as `full_name` and `account_balance` are proportionally represented in validation and test sets.

Analysis of both dataset versions revealed a highly long-tailed distribution: 66% of screens in `pii_v1` contain exactly one PII object, a characteristic that persisted in `pii_v5` despite its threefold increase in size, suggesting it is inherent to the source corpus. This directly influenced the model’s prediction bias as discussed in Section III.

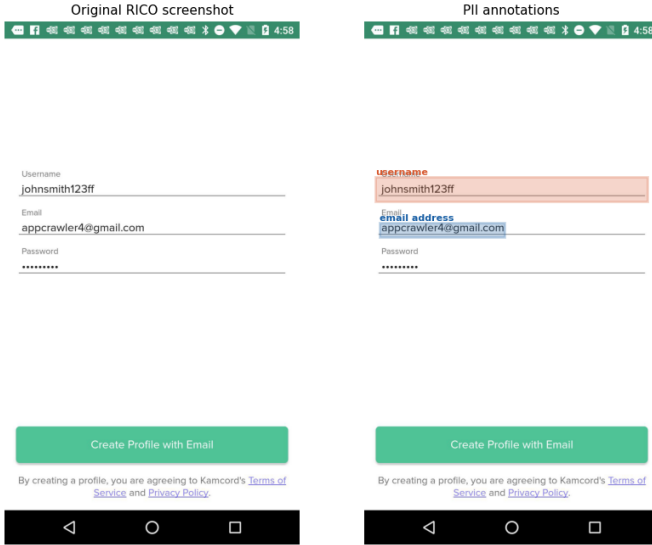


Fig. 1. Example RICO-ScreenQA screen before and after PII annotation. Blue boxes indicate identity/contact labels (`email_address`), orange boxes indicate credential/financial labels (`username`).

III. SINGLE-STAGE VISION-LANGUAGE MODEL

A. Task Formulation

We framed PII detection as autoregressive sequence generation. The model outputs token sequences of the form `<loc0143><loc0512>...sensitive_info`, where location tokens encode normalized coordinates on a 1024×1024 grid.

B. Model and Fine-Tuning

We used PaliGemma 2 [3] (3B parameters, 448-pixel input resolution) in 4-bit NF4 quantization [6]. The vision tower and input embeddings were strictly frozen, LoRA [5] with rank $r = 32$ and scaling $\alpha = 64$ was applied to the language model’s attention and MLP projections. This yielded 47.5 M trainable parameters, or 1.54% of the total.

Evaluation used Intersection over Union (IoU) with Hungarian matching [?] and F1 at the 0.5 IoU threshold.

C. Experiments and Results

Three iterations addressed a structural one-box prediction bias, namely the model rarely predicted more than one bounding box per screen due to the training distribution. Results are reported in Table II.

TABLE II
SINGLE-STAGE VLM RESULTS ON `pii_v1`

Setup	mIoU	P	R	F1
v1 — Baseline	0.570	0.619	0.353	0.449
v2 — Aggressive oversampling ($8\times$)	0.482	0.305	0.399	0.346
v3 — Mild oversampling ($3\times$)	0.491	0.463	0.343	0.394

Data rebalancing alone cannot break the VLM’s autoregressive bias. Aggressive oversampling improved multi-box recall but destroyed precision, as the model began hallucinating boxes on empty screens. The baseline v1 configuration remained the most robust.

IV. TWO-STAGE PIPELINE

Motivated by the one-box bias observed in Section III, we re-architected the task by decoupling localization from classification.

A. Stage 1: OCR Layout Extraction

PaddleOCR [7] (angle-corrected, English) is run on each screenshot to produce a list of tuples containing a bounding box, the recognized text, and a confidence score. No fine-tuning is required.

B. Stage 2: LLM Classifier

We used Gemma-4 [4] (E4B variant) in 4-bit NF4 quantization [6], with LoRA [5] ($r = 32$, $\alpha = 64$) applied to language-model projections only. This yielded 69.8 M trainable parameters, or 0.87% of the total.

Each OCR region is rendered as `[index@x, y] "text"`, with coordinates normalized to a 1000×1000 grid. The model emits a JSON map from tag to PII class.

For training-time label assignment, OCR regions are matched to ground-truth boxes via $\text{IoU} \geq 0.3$ with a containment fallback (at least 70% of the OCR box inside the ground-truth box). A content-based override forces email-shaped and phone-shaped strings to their correct class regardless of the surrounding UI label.

Evaluation reports token-level set precision, recall, and F1 (per class and micro-averaged) along with JSON formatting validity.

C. Experiments and Results

Results across three iterations are shown in Table III.

Per-class F1 scores for v2 trained on `pii_v1` are: `email_address` 0.86, `date_of_birth` 0.74, `full_name` 0.62, `username` 0.58, `phone_number` 0.55, `address` 0.50, `transaction_amount` 0.48, and `account_balance` 0.33.

Per-class F1 scores for v3 trained on `pii_v5` are: `email_address` 0.84, `phone_number` 0.59, `address` 0.56,

TABLE III
TWO-STAGE PIPELINE RESULTS

Setup	Train	P	R	μ F1	MF1
v1 — Flat list, no layout	v1	0.327	0.410	0.364	0.230
v2 — Layout-aware	v1	0.767	0.615	0.683	0.517
v3 — Retrain on pii_v5	v5	0.636	0.544	0.586	0.536

full_name 0.56, account_balance 0.56, transaction_amount 0.55, username 0.47, other_sensitive 0.42, and date_of_birth 0.28.

D. Key Takeaways

Decoupling localization from classification eliminates the one-box bias entirely. The classifier labels every OCR region independently, so dense screens with four or more PII fields are no longer penalized.

Layout matters as much as content. Injecting normalized coordinates into the prompt nearly doubled micro F1 (from 0.36 to 0.68) on the same data, the same model, and the same training budget.

The v3 setup on pii_v5 is a harder evaluation than v2 on pii_v1. The v5 test set contains $6.4\times$ more address and $7.6\times$ more transaction_amount instances - the two hardest classes - than v1. Macro F1 in fact improved from 0.517 to 0.536, and per-class precision rose on account_balance (+0.21), full_name (+0.09), and phone_number (+0.07).

The date_of_birth class regressed sharply (F1 dropped from 0.74 to 0.28) due to label noise: the v5 regex groups date strings (e.g. “03/15/1990”) with age integers (e.g. “25”) under a single label, leaving no consistent signal for the classifier.

The other_sensitive class is structurally weak (F1 = 0.42). It mixes biometrics (weight, height), credentials (password, PIN), and demographics (gender), which share no visual or textual pattern.

Annotation policy mismatch caps recall on address and phone_number. The model correctly declines to label business addresses, phone numbers, and corporate emails as PII, but the ground truth marks all such instances regardless of subject. Reported recall on these classes is therefore a lower bound on real-world performance.

Greedy decoding outperformed beam search. Beam search with a mild length penalty increased entity volume but predominantly wrong ones, dropping micro F1 to 0.607.

V. DEPLOYMENT: SLACK INTEGRATION

To demonstrate real-world applicability, the two-stage pipeline was deployed as a Slack bot (Fig. 2). Users send screenshots to the bot via direct message: a FastAPI backend (running on a Colab GPU, exposed through an ngrok tunnel) runs inference, applies Gaussian blur to detected regions, and returns a preview. The user then approves, manually edits boxes via a browser-based canvas editor, or discards the image. Only the blurred version is posted to the chosen channel. All backend calls are authenticated with a secret API key, and images are processed in memory without persistent storage.

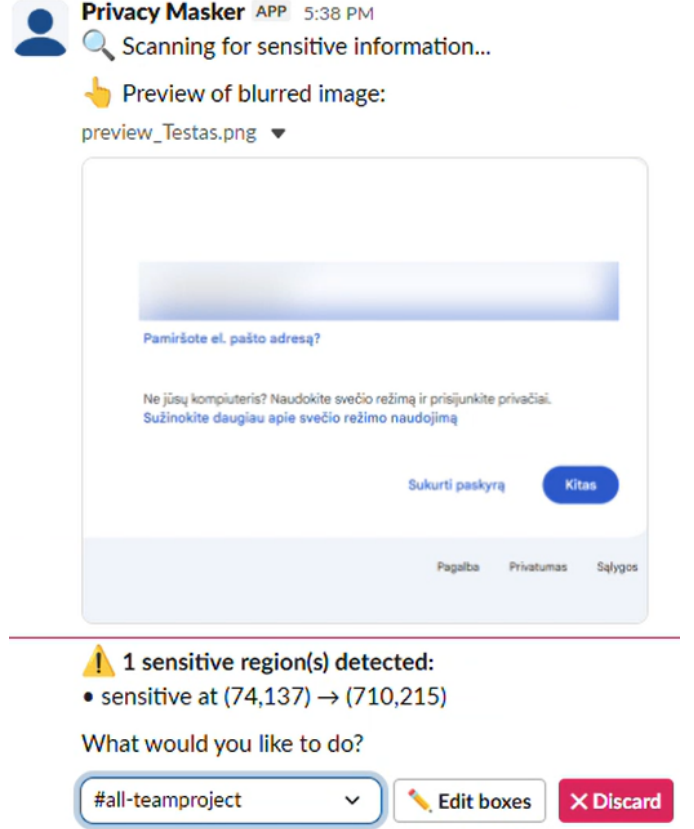


Fig. 2. Blurred preview returned by the Slack bot before publishing.

VI. LIMITATIONS AND FUTURE WORK

The pipeline’s recall is bounded above by the OCR stage. Any text PaddleOCR misses - low-contrast UI elements, text embedded in icons, non-Latin scripts - is invisible to the classifier downstream.

Two concrete label-noise issues were identified in pii_v5. First, date_of_birth conflates date strings with age integers. Second, other_sensitive groups class-incoherent items. Splitting these labels would likely raise micro F1 by 5–10 points without any model changes.

Personal versus business entity ambiguity on address and phone_number is a ground-truth policy issue rather than a model failure. Re-annotating the data with subject-ownership distinctions would raise measured recall on both datasets.

A direct OCR-plus-regex baseline is missing from our evaluation. Such a baseline would quantify how much the LLM classifier contributes beyond simple pattern matching on OCR output.

The single-stage VLM’s one-box bias is a structural constraint, not a training artifact. Higher input resolution (such as PaliGemma’s 896-pixel variant) or grounded-decoding losses might mitigate it, but the two-stage approach sidesteps the problem entirely.

VII. CONCLUSION

We explored two architectures for PII detection on mobile UI screenshots. The single-stage VLM (PaliGemma 2 with LoRA) achieved $F1 = 0.449$ with strong localization ($mIoU = 0.570$), but its structural one-box bias capped recall on dense screens. The two-stage pipeline (PaddleOCR with a Gemma-4 LoRA classifier) reached $F1 = 0.683$ on pii_v1 and 0.586 on pii_v5 (macro $F1 = 0.536$), achieved by decoupling localization from classification and injecting layout-aware coordinates into the prompt.

For small-data, multi-entity detection on structured screens, task decomposition with layout-aware prompting outperforms end-to-end VLM fine-tuning. The v3 evaluation on pii_v5 further isolates label noise - rather than model capacity - as the dominant remaining error source, pointing directly to next steps for data engineering.

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